

Estimation: Astrallys Backend

Stack:

- NestJS
 - Prisma
 - MongoDB
 - REST API
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Modules and Estimated Effort:

1. Setup

Tasks:

- Brainstorming the project structure.
- Preparing the stack.
- Structuring the project.

Complexity: Low **Effort:** 1 day

2. Authentication and Role Management

Technologies: JWT, PassportJS, CaslJS

Tasks:

- Implement user login and registration.
- Configure role-based access control (RBAC).
- Secure API endpoints.

Complexity: Low **Effort:** 1 day

3. CRUD Operations (Users, Tickets, etc.)

Technologies: Prisma

Tasks:

- Create database schemas for users, tickets, and other entities.
- Implement RESTful CRUD APIs.

Complexity: Low **Effort:** 3 days

4. File Management

Technologies: OVH S3

Tasks:

- Integrate OVH S3 for file storage.
- Develop APIs for file upload and retrieval.

Complexity: Low **Effort:** 1 day

5. QR Code Management

Tasks:

- Generate QR codes for tasks or entities.
- Implement QR code validation.

Complexity: Medium **Effort:** 2 days

6. Filtering

Tasks:

- Implement filtering functionality by date, ID, issuer, etc.

Complexity: Low **Effort:** 2 days

7. Real-Time Stats Management

Technologies: Socket.io

Tasks:

- Setup real-time stats updates using WebSockets.
- Create APIs to retrieve and broadcast stats.

Complexity: Low **Effort:** 2 days

CI/CD with GitHub Actions:

Tasks:

- Set up a GitHub repository for the project.
- Create a GitHub Actions workflow for:
 - Building and deploying the project.
 - Linting and code style checks.
- Integrate the pipeline with deployment targets (e.g., staging/production environments).

Complexity: Hard **Effort:** 4 days

Total Estimated Effort: 16 days

Milestones and Timeline:

Milestone	Deliverables	Duration
Setup	Project structure and stack setup	1 day
Authentication	JWT-based auth and RBAC	1 day
CRUD Operations	API implementation for entities	3 days
File Management	S3-based file upload system	1 day

QR Code Management	QR code generation and validation	2 days
Filtering	Add filtering capabilities	2 days
Stats Management	Real-time stats updates	2 days
CI/CD	Deployment	4 days